NGUYEN VIET NGUYEN

AI/ML Engineer

Nguyen Binh Khiem, Dong Hoa, Ho Chi Minh City

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EDUCATION

Bachelor of Artificial Intelligence - Computer Science 2023 - Present
University of Information Technology, VNU-HCM **GPA: 8.59/10.00**

CERTIFICATIONS

- **TOEIC Listening & Reading: 825**/990
- **TOEIC Speaking & Writing: 280**/400
- **AWS Foundation:** AWS Cloud Practitioner Essentials
- **NVIDIA Certificate of Competency:** Applications of AI for Anomaly Detection

PROJECTS

Enhanced FasterViT Architecture for Image Classification Mar 2026 – Jun 2026
Deep Learning / Computer Vision

- Designed FasterViT-Dense by replacing early-stage blocks with custom Dense Capacity Blocks using Conv2d, BatchNorm, GELU, and residual connections to strengthen local feature extraction.
- Improved accuracy by 1.66% to 88.04% and F1-score by 1.64 % on TeaLeafBD, with only 0.5M additional parameters and comparable training time.
- Evaluated FasterViT-0/1/2 variants on CIFAR-10, achieving up to 89.41% test accuracy across multiple model configurations.
- Deployed the optimized model as a Gradio application for real-time classification of seven tea leaf disease categories.

Semantic Poetry Search Engine with Hybrid Retrieval Feb 2026 – May 2026
Information Retrieval / NLP

- Built a hybrid retrieval system combining BM25 and Sentence-BERT (`all-mpnet-base-v2`) to support both keyword-based and semantic poetry search.
- Implemented Reciprocal Rank Fusion to combine sparse and dense rankings, and tuned retrieval parameters with LLM-assisted query evaluation through API on the English Poetry and Cranfield datasets.
- Achieved 0.867 NDCG@10, 0.522 MAP, and 1.000 Recall@10, outperforming the traditional LSI retrieval baseline.
- **Technologies:** Python, NLTK, Sentence-Transformers, Streamlit, Pandas, NumPy.

Vietnamese Social Media Hate Speech Recognition Sep 2025 – Nov 2025
NLP / ML

- Built an end-to-end NLP pipeline to classify Vietnamese social media comments as HATE, OFFENSIVE, or CLEAN, including preprocessing for teencode, abbreviations, and informal language.
- Benchmarked CountVectorizer, TF-IDF, FastText, and PhoBERT representations with Logistic Regression, Naive Bayes, MLP, and BiLSTM classifiers.
- Applied class weighting to address label imbalance; the optimized Logistic Regression model achieved 83.13% accuracy and 61.2% macro F1-score.
- **Technologies:** Python, PyTorch, Scikit-learn, Transformers, FastText, Underthesea.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript
- **AI/ML:** Machine Learning, Deep Learning, Computer Vision, NLP, LLM fundamentals, AI agent
- **Frameworks Libraries:** PyTorch, Scikit-learn, Transformers, LangGraph, OpenCV
- **Tools Deployment:** Git, GitHub, Docker, Gradio, Streamlit, Postman, API Integration